

STEFANOS FOTOPOULOS

Embedded Software Engineer

Athens, Greece | +30 697 190 6645 | stefanos.fotop@gmail.com
linkedin.com/in/stefanos-fotopoulos-95192a290 | github.com/stef-fot

PROFESSIONAL SUMMARY

Embedded Software Engineer with 6 months of professional experience writing, testing and debugging embedded C firmware for STM32 (ARM Cortex-M) microcontrollers. Hands-on with FreeRTOS multithreaded task scheduling, Bluetooth wireless communication, and hardware debugging using oscilloscopes, logic analysers and multimeters. Integrated Master's in Electrical and Computer Engineering, graduated 4th in class. Works collaboratively within a team, communicates clearly in written and spoken English, and is motivated to build a long-term career in embedded software engineering.

PROFESSIONAL EXPERIENCE

Embedded Software Engineer

February 2025 - July 2025

IQSoft Greece, Athens, Greece

- Wrote, tested and documented embedded C firmware for STM32 (ARM Cortex-M) microcontrollers, delivering features across 3 product modules from requirement through to hardware validation.
- Debugged firmware and hardware issues using an oscilloscope, logic analyser and multimeter, diagnosing timing, signal integrity and protocol faults across I2C, SPI and UART interfaces.
- Implemented multithreaded task scheduling and prioritisation on FreeRTOS, reducing worst-case latency in time-critical control loops to under 20 ms.
- Developed Bluetooth wireless serial communication for real-time data exchange between embedded devices and host applications, sustaining stable throughput at 100 kbps.
- Designed and tuned closed-loop control algorithms for stepper and servo motors, improving positioning accuracy and reducing settling time by 20%.
- Wrote unit tests and applied structured testing practices during hardware bring-up, shortening the debug cycle by 30%.
- Worked closely with senior engineers in an Agile workflow, managing code in Git and reporting progress through regular team reviews.

Software Engineer (Internship)

July 2024 - August 2024

Neunen, Athens, Greece

- Built and maintained application features using React, TypeScript and Java, following established coding and code review practices.
- Implemented backend services and RESTful APIs, then validated and documented endpoints with Postman to catch integration defects before release.

PROJECTS

USB-C Power Delivery Controller — Diploma Thesis

Embedded C, STM32G474

- Implemented USB-C Power Delivery contract negotiation firmware between source and sink devices in embedded C, enabling dynamic voltage selection from user input.
- Designed event-driven firmware with real-time voltage and current monitoring and power safety mechanisms preventing over-voltage and over-current conditions; validated behaviour with MATLAB/Simulink simulation before hardware bring-up.

Emotion Detector — Multilingual NLP Application

Python, Flask

- Built an end-to-end machine learning application classifying sentiment and emotion across multiple languages, served through a Flask REST backend with a responsive web interface.

Thesis Management System

Spring Boot, MySQL

- Developed a full-stack web application with role-based authentication (Spring Security), strategy-based assignment workflows and a Thymeleaf front end backed by MySQL.

TECHNICAL SKILLS

Programming Languages: C, Embedded C, C++, Python, MATLAB, VHDL, Java, JavaScript, TypeScript, SQL

Embedded Systems: STM32 (ARM Cortex-M), FreeRTOS (multithreaded micro-kernel RTOS), bare-metal and HAL/LL firmware, interrupts, timers, PWM, ADC, DMA, GPIO, motor control (stepper and servo), USB-C Power Delivery

Protocols & Wireless: Bluetooth, I2C, SPI, UART, USB-C PD, REST APIs

Debugging & Test Tools: Oscilloscope, logic analyser, multimeter, protocol analysis, SWD/JTAG debugging, unit testing, hardware bring-up and troubleshooting

Development Practices & Tools: Agile, code review, unit and integration testing, Git, GitHub, STM32CubeIDE, STM32CubeMX, Linux, Postman, AI-assisted development tools

Other: Machine learning, computer vision, MATLAB/Simulink simulation, Spring Boot, Flask, React, MySQL

EDUCATION

University of Ioannina, Greece

September 2019 - February 2025

Integrated Master's Degree (5-year Diploma), Electrical and Computer Engineering

- GPA: 7.4 / 10 — ranked 4th in class. Thesis: USB-C Power Delivery negotiation firmware on the STM32G474.
- Relevant coursework: Embedded Systems, Digital Design (VHDL), Control Systems, Signals and Systems, Computer Architecture, Machine Learning.

CERTIFICATIONS & LANGUAGES & OTHERS

Certifications: IBM Generative AI Engineering (June 2026) | MathWorks Computer Vision Engineer (Mar 2026) | MathWorks Medical Image Processing (Apr 2026)

Languages: Greek (Native) | English (Fluent, C1) | German (Intermediate, A2)

Military Service: Completed | Sep 2025 - Sep 2026